



Department of Computer Science
COMP 4381-Data Science and Analysis

Flight Delay Machine learning Model Phase Report

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Introduction

This is the final phase of our flight delay prediction project. We built a machine learning model to predict whether an airport-day record would be classified as delayed or on time. The delay threshold is **15 minutes** average schedule delay.

What we did:

- Loaded the cleaned dataset from Phase 3
- Preprocessed the data using a ColumnTransformer pipeline
- Trained a **Logistic Regression** model
- Evaluated model performance on test data
- Checked for overfitting/underfitting

- Visualized delay patterns by country and season

Preprocessing Pipeline

We used a **ColumnTransformer**

Numeric Pipeline:

- `SimpleImputer(strategy='mean')` → fills missing values with mean
- `StandardScaler()` → scales features (mean = 0, std = 1)

Categorical Pipeline:

- `SimpleImputer(strategy='most_frequent')` → fills missing with most common category
- `OneHotEncoder()` → converts categories to binary columns

This pipeline was wrapped into the model using `make_pipeline()`.

Model: Logistic Regression

Why Logistic Regression?

- Standard binary classification algorithm taught in the course
- Uses **sigmoid function** to output probabilities between 0 and 1
- Simple, interpretable, and works well for this problem

How it works :

The model calculates the probability that a record is delayed:

$$P(\text{Delayed}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}} \quad P(\text{Delayed}) = 1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}$$

If probability > 0.5, predict **Delayed (1)**.

If probability < 0.5, predict **On Time (0)**.

Result :

Confusing Matrix :

	Predicted On Time (0)	Predicted Delayed (1)
Actually on Time (0)	4223	416
Actually Delayed (1)	1704	3769

Performance Metrics :

Metric	Value
Accuracy	74.25%
Recall	68.93%
Precision	73.33%
F1-Score	71.07%

Interpretation:

- **Accuracy (74%):** The model is correct about 3 out of 4 times
- **Recall (69%):** Catches 69% of actual delayed days
- **Precision (73%):** When predicting delay, correct 73% of the time
- **F1-Score (71%):** Balanced measure of precision and recall

Classification Report :

Class	Precision	Recall	F1-Score	Support
On Time (0)	0.75	0.91	0.82	4639
Delayed (1)	0.73	0.69	0.71	5473

Overfit/Underfit Check :

Metric	Train	Test	Gap
Accuracy	74.39%	74.25%	0.14%

Verdict: No overfitting or underfitting detected. The model generalises well.

Visualizations

Delay Rate by Country

Country	Delay Rate	Record
Netherlands	54%	1931
Germany	53.8%	7539
France	52.6%	8899
Spain	52.6%	4082
UK	51.5%	11953
Italy	51.1%	6684

Observation: Delay rates are fairly consistent across countries, ranging from 51% to 54%.

Delay Rate by Season

Observation: Summer and Winter show higher delay counts, while Spring and Fall show relatively lower delays. This likely reflects weather impacts and passenger volume fluctuations.

Discussion

What the Model Tells Us

The Logistic Regression model achieved **74% accuracy**, which is a meaningful improvement over random guessing (~50%). Key insights:

1. **The model is better at predicting on-time days** (91% recall) than delayed days (69% recall). This means it's more conservative about predicting delays.
2. **Country-level differences** exist but are relatively small, suggesting that operational factors at the airport level may be more important than country-level factors.
3. **Seasonal patterns** indicate that weather and volume play a role in delays.
4. **Preprocessing pipeline worked well**, handling missing values and scaling features correctly.

What We Learned

- **Data preprocessing is critical** — proper handling of missing values and scaling makes a big difference
- **Simple models can work well** — Logistic Regression provides a strong baseline
- **Always check for overfitting** — comparing train and test performance is essential
- **Visualizations add context** — country and seasonal patterns help explain model behavior

Conclusion :

This project successfully built a complete data science pipeline:

1. Loaded cleaned data from Phase 3
2. Defined binary target `Is_Delayed`
3. Built preprocessing pipeline with `ColumnTransformer`
4. Trained Logistic Regression model
5. Evaluated using accuracy, recall, precision, F1-score, and confusion matrix
6. Checked for overfitting/underfitting
7. Visualized delay patterns by country and season

Key takeaway: Airport-level delay patterns can be predicted from historical punctuality data. Logistic Regression provides an interpretable, effective baseline model for this binary classification task.